

Final Assignment 18.1: Platforms for Handling Big Data

Part 1: Ingesting Data Into the HDFS

1.

```
C:\tmp\emeritus\testprogram>dir
Volume in drive C is OS
Volume Serial Number is 3C4C-CF74

Directory of C:\tmp\emeritus\testprogram

12/20/2021  04:47 PM    <DIR>          .
12/20/2021  04:47 PM    <DIR>          ..
12/20/2021  04:38 PM                45 Manifest.txt
05/05/2014  02:58 PM            1,367 SalesCountryDriver.java
05/05/2014  02:58 PM            749 SalesCountryReducer.java
05/05/2014  02:58 PM       123,637 SalesData.csv
05/05/2014  02:58 PM            659 SalesMapper.java
           5 File(s)        126,457 bytes
           2 Dir(s)    326,070,628,352 bytes free
```

2.

```
C:\tmp\emeritus>docker cp testprogram/ namenode:/home

C:\tmp\emeritus>
```

3.

```
root@30b6d440d7c2:/home/testprogram# hdfs dfs -copyFromLocal SalesData.csv /inputMapReduce
2021-12-22 02:57:40,968 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
root@30b6d440d7c2:/home/testprogram#
```

4.

```
root@30b6d440d7c2:/home/testprogram# hdfs dfs -cat /inputMapReduce/SalesData.csv
2021-12-22 02:58:26,872 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
1/2/09 4:53,Product1,1200,Visa,Betina,Parkville,MO,United States,1/2/09 4:42,1/2/09 7:49,39.195,-94.6
1/2/09 13:08,Product1,1200,Mastercard,Federica e Andrea,Astoria,OR,United States,1/1/09 16:21,1/3/0
1/4/09 12:56,Product2,3600,Visa,Gerd W ,Cahaba Heights,AL,United States,11/15/08 15:47,1/4/09 12:45,33.52
1/4/09 13:19,Product1,1200,Visa,LAURENCE,Mickleton,NJ,United States,9/24/08 15:19,1/4/09 13:04,39.79
1/4/09 20:11,Product1,1200,Mastercard,Fleur,Peoria,IL,United States,1/3/09 9:38,1/4/09 19:45,40.6
1/2/09 20:09,Product1,1200,Mastercard,adam,Martin,TN,United States,1/2/09 17:43,1/4/09 20:01,36.3
1/4/09 13:17,Product1,1200,Mastercard,Renee Elisabeth,Tel Aviv,Tel Aviv,Israel,1/4/09 13:03,1/4/09 22:10,32.0666667,34.7
1/5/09 2:42,Product1,1200,Diners,Stacy,New York,NY,United States,1/5/09 2:23,1/5/09 4:59,40.71417,-7
1/2/09 9:16,Product1,1200,Mastercard,Sean ,Shavano Park,TX,United States,1/2/09 8:32,1/5/09 9:05,29.423
1/5/09 10:08,Product1,1200,Visa,Georgia,Eagle,ID,United States,11/11/08 15:53,1/5/09 10:05,43.69
1/2/09 14:18,Product1,1200,Visa,Richard,Riverside,NJ,United States,12/9/08 12:07,1/5/09 11:01,40.032
1/2/09 7:35,Product1,1200,Diners,Hani,Salt Lake City,UT,United States,12/30/08 5:44,1/6/09 10:52,40.76083
1/5/09 4:10,Product1,1200,Mastercard,Nicola Boenderoot,Gauteng,South Africa,1/5/09 2:33,1/7/09 5:13,-26.1666667,27.8666
```

Part 2: Performing MapReduce: Aggregation Sales by Country

1.

`SalesCountryDriver.java`

This file is the Job Creator which controls the *flow* of the data processing.

`SalesMapper.java`

This file is used for splitting records for distributed calculation. The input data is split into smaller *blocks* and each of them is assigned to a mapper for processing.

`SalesCountryReducer.java`

This file is used for processing *chunks* and combining output. The Reduce task takes the output from the Map as an input.

2.

```
root@30b6d440d7c2:/home/testprogram# $JAVA_HOME
bash: /usr/lib/jvm/java-8-openjdk-amd64/: Is a directory
root@30b6d440d7c2:/home/testprogram#
```

3. The code should appear exactly as below with no errors:

```
javac -d . SalesMapper.java SalesCountryReducer.java
SalesCountryDriver.java
```

```
root@30b6d440d7c2:/home/testprogram# export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/jre/
root@30b6d440d7c2:/home/testprogram# export CLASSPATH="$HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-client-core-3.2.1.jar:$HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-client-common-3.2.1.jar:$HADOOP_HOME/share/hadoop/common/hadoop-common-3.2.1.jar:~/testprogram/SalesCountry/*:$HADOOP_HOME/lib/*"
root@30b6d440d7c2:/home/testprogram# export HDFS_NAMENODE_USER="root"
root@30b6d440d7c2:/home/testprogram# export HDFS_DATANODE_USER="root"
root@30b6d440d7c2:/home/testprogram# export HDFS_SECONDARYNAMENODE_USER="root"
root@30b6d440d7c2:/home/testprogram# export YARN_RESOURCEMANAGER_USER="root"
root@30b6d440d7c2:/home/testprogram# export YARN_NODEMANAGER_USER="root"
root@30b6d440d7c2:/home/testprogram# javac -d . SalesMapper.java SalesCountryReducer.java SalesCountryDriver.java
root@30b6d440d7c2:/home/testprogram#
```

4. The code should appear exactly as below with no errors:

```
jar cfm ProductSalePerCountry.jar Manifest.txt
SalesCountry/*.class
```

```
root@30b6d440d7c2:/home/testprogram# jar cfm ProductSalePerCountry.jar Manifest.txt SalesCountry/*.class
root@30b6d440d7c2:/home/testprogram#
```

5. The code should appear exactly as below with no errors:

```
hadoop jar ProductSalePerCountry.jar /inputMapReduce
/mapreduce_output_sales
```

```
root@30b6d440d7c2:/home/testprogram# hadoop jar ProductSalePerCountry.jar /inputMapReduce /mapreduce_output_sales
2021-12-22 19:35:03,304 INFO client.RMProxy: Connecting to ResourceManager at resourcemanager/172.19.0.3:8032
2021-12-22 19:35:03,420 INFO client.AHSProxy: Connecting to Application History server at historyserver/172.19.0.4:10200
2021-12-22 19:35:03,445 INFO client.RMProxy: Connecting to ResourceManager at resourcemanager/172.19.0.3:8032
2021-12-22 19:35:03,446 INFO client.AHSProxy: Connecting to Application History server at historyserver/172.19.0.4:10200
2021-12-22 19:35:03,573 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement
the Tool interface and execute your application with ToolRunner to remedy this.
2021-12-22 19:35:03,590 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/
root/.staging/job_1640178088064_0001
2021-12-22 19:35:03,668 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteh
ostTrusted = false
2021-12-22 19:35:03,776 INFO mapred.FileInputFormat: Total input files to process : 2
2021-12-22 19:35:03,796 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteh
ostTrusted = false
2021-12-22 19:35:03,814 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteh
ostTrusted = false
2021-12-22 19:35:04,235 INFO mapreduce.JobSubmitter: number of splits:2
2021-12-22 19:35:04,317 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteh
ostTrusted = false
2021-12-22 19:35:04,328 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1640178088064_0001
2021-12-22 19:35:04,328 INFO mapreduce.JobSubmitter: Executing with tokens: []
2021-12-22 19:35:04,475 INFO conf.Configuration: resource-types.xml not found
2021-12-22 19:35:04,475 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2021-12-22 19:35:04,778 INFO impl.YarnClientImpl: Submitted application application_1640178088064_0001
2021-12-22 19:35:04,812 INFO mapreduce.Job: The url to track the job: http://resourcemanager:8088/proxy/application_1640
178088064_0001/
2021-12-22 19:35:04,813 INFO mapreduce.Job: Running job: job_1640178088064_0001
2021-12-22 19:35:10,921 INFO mapreduce.Job: Job job_1640178088064_0001 running in uber mode : false
2021-12-22 19:35:10,921 INFO mapreduce.Job: map 0% reduce 0%
2021-12-22 19:35:15,967 INFO mapreduce.Job: map 50% reduce 0%
```

6. The code should appear exactly as below with no errors:

```
hdfs dfs -cat /mapreduce_output_sales/part-00000
```

```
root@30b6d440d7c2:/home/testprogram# hdfs dfs -cat /mapreduce_output_sales/part-00000
2021-12-22 19:36:16,249 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteh
ostTrusted = false
Argentina      2
Australia     76
Austria 14
Bahrain 2
Belgium 16
Bermuda 2
Brazil 10
Bulgaria      2
CO            2
Canada 152
Cayman Isls   2
China 2
Costa Rica    2
Country 2
Czech Republic 6
Denmark 30
Dominican Republic 2
Finland 4
France 54
Germany 50
```